

EXNO: 2

Analysis of SWR (Standing Wave Ratio) Transmission Lines by varying a) load impedance b) characteristic impedance

TEST CASE 1

Analyze the SWR of a transmission line for a fixed characteristic impedance (Z_0) of 50 ohms, as the load impedance (Z_L) is varied across 650 ohms to 850 ohms. Verify how the SWR changes with each value of Z_L . When $Z_L = Z_0$, the SWR should equal 1 (indicating no reflection).

OUTPUT:

$Z_0 = 50$ ohms Variable: Z_R

S.No	(Z_L) (Ω)	Simulation SWR	Theoretical SWR
1	650	19.96	13.00
2	700	20.46	14.00
3	750	21.03	15.00
4	800	21.65	16.00
5	850	22.32	17.00

$Z_R = 50 / 40$ ohms Variable: Z_0

ZO Ohms	Simulation results	Theoretical results
	SWR	SWR
650	2.04	1
700	1.94	1
750	1.86	1
800	1.79	1
850	1.73	1

Simulation Results:

Scilab

code:

```
1 clc;clear;close;
2 Zo=50;ZR=650-(%i*475);
3 K=(ZR-Zo)/(ZR+Zo);
4 ampK=sqrt((real(K)^2)+
(imag(K)^2));S=(1+ampK)/(1-ampK);
5 printf("(c)Standing wave ratio =
%f",round(S*1000)/1000);
```

Result:

(c)Standing wave ratio = 19.969000

Execute

Reset

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Scilab

code:

```
1 clc;clear;close;
2 Zo=650;ZR=650-(%i*475);
3 K=(ZR-Zo)/(ZR+Zo);
4 ampK=sqrt((real(K)^2)+
(imag(K)^2));S=(1+ampK)/(1-ampK);
5 printf("(c)Standing wave ratio =
%f",round(S*1000)/1000);
```

Result:

(c)Standing wave ratio = 2.045000

Execute

Reset

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Theoretical Results:

Given

$$Z_L = 650\Omega \quad \Gamma = \frac{650-50}{650+50} = 0.857$$

$$SWR = \frac{1+0.857}{1-0.857} = 13$$

$$Z_L = 700\Omega \quad \Gamma = \frac{700-50}{700+50} = 0.8667$$

$$SWR = 14$$

$$Z_L = 750\Omega \quad \Gamma = \frac{750-50}{750+50} = 0.875 \quad SWR = 15$$

$$Z_L = 800\Omega \quad \Gamma = \frac{800-50}{800+50} = 0.8824 \quad SWR = 16$$

$$Z_L = 850\Omega \quad \Gamma = \frac{850-50}{850+50} = 0.88 \quad SWR = 17$$

$Z_L = Z_0$ then,

$$\Gamma = \frac{650-650}{650+650} = 0$$

$$SWR = \frac{1+0}{1-0} = 1$$

for all $SWR = 1$

Test Case 2:

Analyze the SWR for a fixed load impedance (Z_L) of 50 ohms while varying the characteristic impedance (Z_0) of the transmission line to 250 ohms to 400 ohms. Verify how the SWR changes with each value Z_0 including $Z_0=Z_L$.

OUTPUT:

$Z_R=50/_40\text{o}$ ohms Variable: Z_0

S.No	(Z_0) (Ω)	Simulation SWR	Theoretical SWR
1	50	1.00	92.23
2	250	5.00	23.20
3	300	6.00	21.16
4	350	7.00	19.98
5	400	8.00	19.35

Simulation Results:

Scilab

code:

```
1 clc;clear;close;
2 Zo=250;ZR=50-(%i*475);
3 K=(ZR-Zo)/(ZR+Zo);
4 ampK=sqrt((real(K)^2)+(imag(K)^2));S=(1+ampK)/(1-ampK);
5 printf("(c)Standing wave ratio = %f",round(S*1000)/1000);
```

Result:

```
(c)Standing wave ratio = 23.207000
```

Execute ⚙️ Reset

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Theoretical Results

Handwritten calculations for Standing Wave Ratio (SWR) based on impedance mismatch:

$Z_L = 50$, $Z_0 = 250 - 400j$

$Z_0 = 50 \Rightarrow \Gamma = \frac{50-50}{50+50} = 0$ SWR = 1

$Z_0 = 250 \quad \Gamma = \frac{50-250}{50+250} = -0.66$ SWR = 5

$Z_0 = 300 \quad \Gamma = \frac{50-300}{50+300} = -0.71$ SWR = 6

$Z_0 = 350 \quad \Gamma = \frac{50-350}{50+350} = -0.75$ SWR = 7

$Z_0 = 400 \quad \Gamma = \frac{50-400}{50+400} = -0.77$ SWR = 8